

Before you use the FSDB API of Debussy 5.3

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Important Changes

FSDB Writer API of Debussy 5.3 can dump much compact simulation results. Typically, it gets file size 20% (for VHDL design) ~ 50% (for Verilog designs) saved.

For the new FSDB format and if you had developed FSDB reader(s) through Novas's older FSDB Reader API, you have to check the usage on **ffrGetVC()** to avoid errors.

When you use **ffrGetVC()** of FSDB Reader API of Debussy 5.3 to get value changes of a signal, please remember to copy returned value to you own buffer or use it immediately. DON'T store the returned pointer for further usage. We don't guarantee that kind of pointers will always point to the correct value change.

For example, if an integer signal has value changes at

<i>Time</i>	<i>value</i>
32	(0 0)
4567890	(0 20)

If you have the following code segment

```
time.H = 0;
time.L = 0;
vc_trvs_hdl->ffrGotoXTag(&time);
vc_trvs_hdl->ffrGetVC(&vc_ptr);
fprintf(stdout, "1.           [Use It Immediately] Value at (0 0) is %d\n",
    *((int*)vc_ptr));
// Copy to a local buffer
memcpy((void*) &int_val1, (void*) vc_ptr, sizeof(int));

time.H = 0;
```

```

time.L = 10;

vc_trvs_hdl->ffrGotoXTag(&time);
vc_trvs_hdl->ffrGetVC(&vc_ptr);
// Copy to a local buffer
memcpy((void*) &int_val2, (void*) vc_ptr, sizeof(int));

fprintf(stdout, "1.          [Use It Immediately] Value at (0 10) is %d\n",
        *((int*)vc_ptr));
fprintf(stdout, "2.  [Store the Value and Use It Later] Value at (0 0) is %d\n",
        int_val1);
fprintf(stdout, "2.  [Store the Value and Use It Later] Value at (0 10) is %d\n",
        int_val2);
fprintf(stdout, "3. [Store the Pointer and Use It Later] Value at (0 0) is %d\n",
        *((int*)vc_ptr));
fprintf(stdout, "3. [Store the Pointer and Use It Later] Value at (0 10) is %d\n",
        *((int*)vc_ptr));

```

Then it will produce the result as below

1. [Use It Immediately] Value at (0 0) is 32
1. [Use It Immediately] Value at (0 10) is 4567890
2. [Store the Value and Use It Later] Value at (0 0) is 32
2. [Store the Value and Use It Later] Value at (0 10) is 4567890
3. [Store the Pointer and Use It Later] Value at (0 0) is 4567890
3. [Store the Pointer and Use It Later] Value at (0 10) is 4567890

You'll find that if you use it immediately(1) or copy the value to your own buffer and use it later(2), both the results are correct. However, if you store the pointer and use it later(3), you only get the last value change which is not the expected result.